6 **(a)** Fig. 6.1 shows some energy levels for the hydrogen atom (not to scale).

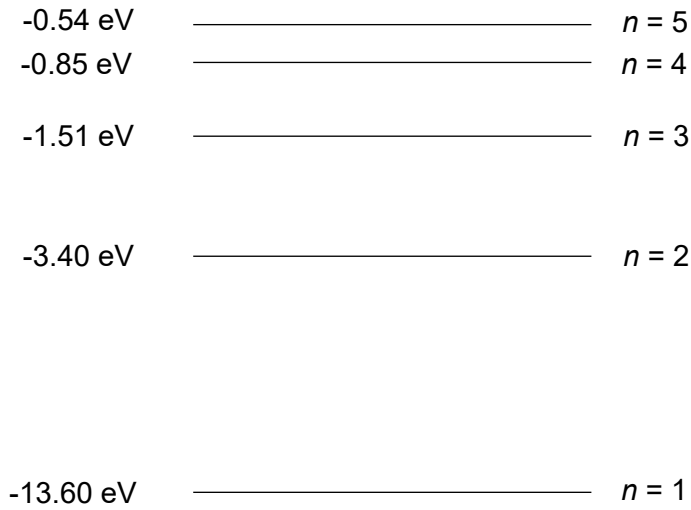

Fig 6.1

(i) State the number of emission lines that could be produced from the energy levels shown in Fig 6.1.

number of emission lines = …… [1]

Visible light wavelengths range approximately from 400 nm to 750 nm.

(ii) Show that the emission spectrum when electrons make transitions to the $n = 1$ state does not fall within the visible region of the electromagnetic spectrum.

[3]

(iii) Hence or otherwise, determine the number of *visible* emission lines for the energy levels shown in Fig. 6.1. Show all workings clearly.

number of visible emission lines = …… [4]

(b) The atomic energy levels for a multi-electron atom can be approximated using the formula:

$$E_n = -\frac{(13.60)(Z-1)^2}{n^2}$$

where Z is the atomic number of the atom, n is the quantum number of the shell, and E_n is the energy of the shell in eV.

(i) Using the formula provided in **(b)**, determine the energy of the photon released when an electron transitions from $n = 2$ to $n = 1$ in a tungsten atom (atomic number 74).

energy of photon released = ... eV [3]

(ii) Hence state the type of electromagnetic radiation emitted when an electron transitions from $n = 2$ to $n = 1$ in a tungsten atom.

..... [1]